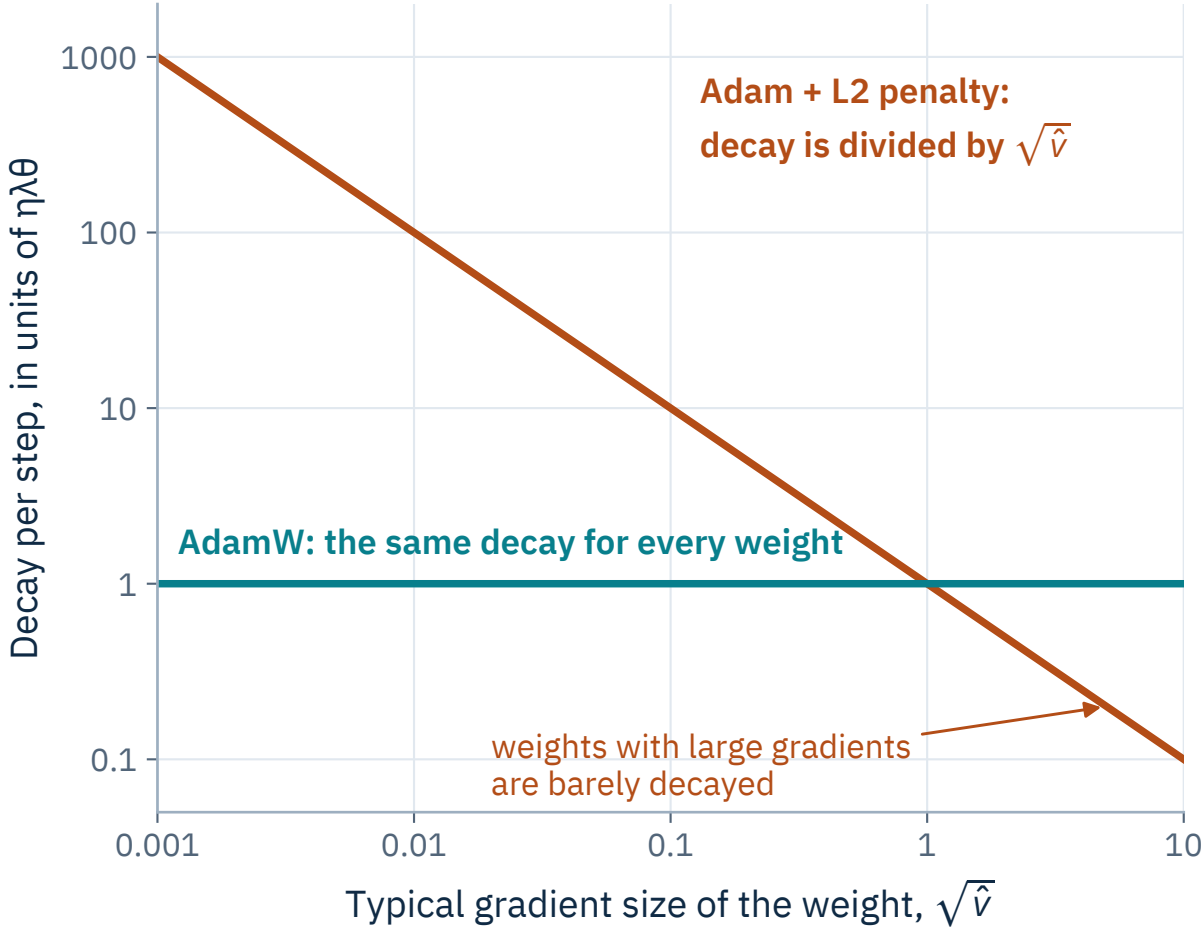
Adam + L2 penalty:
decay is divided by $\sqrt{\hat{v}}$
AdamW: the same decay for every weight
weights with large gradients
are barely decayed
Decay per step, in units of ηλθ
1000
100
10
1
0.1
0.001
0.01
0.1
1
10
Typical gradient size of the weight, $\sqrt{\hat{v}}$